

**VISVESVARAYA TECHNOLOGICAL UNIVERSITY
BELAGAVI – 590 018.**



A Internship Report

On

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Submitted in Partial Fulfilment of the Requirement for

The Award of the Degree Of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE & ENGINEERING

Submitted by,

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2025-2026

**VISVESVARAYA TECHNOLOGICAL UNIVERSITY
BELAGAVI – 590 018.**



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CERTIFICATE OF APPROVAL OF INTERNSHIP

This is to certify that **Ishwari Hallur** bearing **USN:2KD22CS034** has satisfactorily completed the Internship entitled “**EduTech Learning Platform**” in **Bharati Unnati AI-Fellowship** for the partial fulfilment of Bachelor of Engineering in Computer Science and Engineering prescribed by the **Visvesvaraya Technological University, Belagavi** for the academic year 2025-26. The report has been approved as it satisfies the academic requirements in Internship prescribed for the Bachelor of Engineering Degree.

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ABSTRACT

The **Bharat Unnati AI Fellowship** internship provided hands-on exposure to emerging AI technologies, structured around Generative AI, Agentic AI, and practical software development. The project developed an **AI-Based Workflow Automation System**, which integrates prompt-engineered Generative AI models (e.g. GPT-4) with an automation workflow engine (n8n) to process tasks autonomously. Key tasks included designing prompts for summarization, classification, and code generation, building n8n workflows to chain AI calls and external APIs, and evaluating system outputs. Coding fundamentals were reinforced by solving ~300 algorithmic problems, strengthening logical thinking and implementation skills.

Throughout the internship, extensive use was made of modern AI frameworks and tools. For example, state-of-the-art LLMs (large language models) such as GPT-4 were used for content generation and summarization tasks. Prompt engineering was crucial for guiding the AI models to produce the desired outputs. The system's architecture leveraged a Transformer-based LLM (encoder-decoder) as the core language model. The workflow automation engine (n8n) orchestrates the pipeline, integrating the AI model with other services (databases, web APIs). By the end of the internship, a functional AI-driven workflow prototype was built, demonstrating how multi-step tasks ("summarize this article and classify its topic") can be executed autonomously.

In summary, this internship bridged theory and practice: it showed how modern generative models can automate content-related tasks, and how to incorporate them in scalable workflows. The hands-on experience with AI tools, cloud services, and programming solidified technical knowledge and prepared for future AI projects and careers.

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CHAPTER 1

INTRODUCTION

1.1 Introduction

In today's rapidly evolving technological landscape, Artificial Intelligence (AI) has become one of the most influential fields shaping innovation across industries. From automating repetitive tasks to enabling intelligent decision-making, AI is transforming the way systems are designed and operated. As a student aiming to build a career in this domain, gaining practical exposure beyond classroom learning is essential. This internship, undertaken as part of the “**Bharat Unnati AI Fellowship**” provided an opportunity to explore real-world applications of AI and strengthen both technical and professional competencies.

The internship was conducted with **Learners Byte** in collaboration with **Global Info vision**. The program was structured as a virtual, project-oriented training experience, allowing participants to work in a flexible learning environment while engaging with industry-relevant tools and concepts. The internship site primarily consisted of an online learning and development platform where resources such as recorded sessions, assignments, and practical tasks were provided. Regular guidance and structured modules ensured a systematic approach to learning, simulating a professional work environment.

The main objective of this internship was to bridge the gap between theoretical knowledge and its practical implementation. The fellowship introduced multiple domains—**Cod expert**, **Generative AI**, **Agentic AI** each contributing to a different aspect of skill development. The work carried out during the internship involved hands-on exercises, tool-based learning, and task-oriented assignments that required active

problem-solving and critical thinking.

1.2 Significance

A significant portion of the internship work was focused on **Generative AI**, where tools such as ChatGPT and Gemini were used extensively. Through these platforms, tasks such as prompt design, AI-assisted content generation, and problem-solving were performed. This helped in understanding how AI models interpret inputs and generate meaningful outputs.

The experience also highlighted the importance of prompt engineering, where even small changes in input can significantly influence the results produced by AI systems. This domain provided valuable insights into how generative models are being applied in areas like content creation, coding assistance, and knowledge retrieval.

Another important component of the internship was **Agentic AI**, which focused on building and understanding autonomous AI systems. For this domain, the tool n8n was utilized to design and implement automated workflows. This involved creating step-by-step processes where different nodes were connected to perform tasks such as data handling, API integration, and decision-making operations. Working with n8n provided a clear understanding of how AI agents can be structured to perform complex, multi-step activities with minimal human intervention. It also demonstrated the practical application of automation in improving efficiency and reducing manual effort.

The **Code Expert** domain supported the development of programming and logical reasoning skills. The work in this area involved solving coding problems, understanding basic algorithms, and improving problem-solving techniques. This domain acted as a foundation, enabling better implementation of concepts learned in AI-related modules.

1.3 Generative AI (Gen-AI)

Generative Artificial Intelligence (Gen-AI) is a powerful branch of AI that focuses on creating new and original content such as text, images, audio, video, and even computer code. Unlike traditional AI systems that primarily analyze or classify data, generative AI produces new outputs by learning patterns from large datasets.

Generative AI models are typically based on deep learning techniques, especially **Large Language Models (LLMs)**, which are trained on vast amounts of data. These models can generate human-like responses based on user inputs, making them highly useful in various applications.

The working of generative AI involves multiple stages, including data collection, model training, and content generation. When a user provides a prompt, the AI processes the input and generates a response based on learned patterns and probabilities.

Generative AI is widely used in content creation, chatbot development, code generation, and educational platforms. It helps in automating tasks, improving productivity, and enhancing creativity. However, it also has limitations such as generating incorrect information and requiring human supervision for validation.

1.4 CodeXpert (AI in Coding)

CodeXpert refers to AI-powered coding systems that assist developers in writing, debugging, and optimizing code efficiently. It represents the integration of generative AI into software development, making coding faster and more accurate.

These systems can understand programming languages and generate code based on user requirements. They also provide features such as auto-completion, error detection, debugging suggestions, and code optimization.

The working process of CodeXpert involves taking user input in the form of a coding problem or requirement, analyzing it using AI models, and generating relevant code solutions. Developers can then review, modify, and implement the generated code.

CodeXpert plays a significant role in modern software development and education. It helps beginners learn programming concepts easily while enabling experienced developers to improve productivity. However, the generated code must always be verified to ensure correctness and security.

1.5 Agentic AI

Agentic AI represents the next stage in the evolution of artificial intelligence, focusing on autonomous decision-making and task execution. Unlike generative AI, which primarily generates content, agentic AI systems can plan, act, and achieve specific goals with minimal human intervention.

Agentic AI operates through a continuous cycle of perception, planning, execution, and learning. It understands the environment, makes decisions, performs actions, and improves based on feedback. This makes it highly effective for handling complex and multi-step tasks.

These systems are widely used in automation, smart assistants, workflow management, and advanced applications such as robotics and intelligent systems. In educational platforms, agentic AI can act as a smart tutor, track student progress, and provide personalized learning paths.

Although agentic AI offers significant advantages in terms of efficiency and automation, it also presents challenges such as ethical concerns, system complexity, and the need for proper monitoring.

1.6 Integration of Technologies in EdTech

The combination of Generative AI, CodeXpert, and Agentic AI plays a crucial role in enhancing modern educational platforms. Generative AI helps in creating study materials, quizzes, and explanations. CodeXpert assists students in learning programming through real-time coding support. Agentic AI enables personalized learning by analyzing student performance and adapting learning paths accordingly.

CHAPTER 2

SYSTEM REQUIREMENT SPECIFICATION AND ANALYSIS

2.1 Introduction

System Requirement Specification (SRS) is an essential document in the software development lifecycle that clearly defines the requirements, functionalities, and constraints of a system. It acts as a blueprint for developers, designers, and stakeholders to understand how the system should behave and what features it must include.

In this project, the SRS is designed for an **AI-based EdTech Learning Platform** that integrates advanced technologies such as **Generative AI, CodeXpert, and Agentic AI**. The purpose of this platform is to enhance the learning experience by providing intelligent content, coding assistance, and automated workflows.

This chapter focuses on identifying both functional and non-functional requirements, analyzing system feasibility, and describing the overall system architecture. It ensures that the system is well-structured, efficient, and capable of meeting user needs effectively.

2.2 System Overview

The proposed system is a web-based educational platform that enables students to learn courses interactively and efficiently. It is designed to provide a personalized learning experience by utilizing AI technologies.

The platform allows users to:

- Register and log in securely
- Select and enroll in courses
- Access learning materials
- Take quizzes with a timer
- Receive instant feedback and results

The system also includes AI-based features such as:

- Content generation using Generative AI

- Coding support using CodeXpert
- Automated task handling using Agentic AI

The use of **n8n workflow automation** helps in connecting different modules and automating processes such as quiz evaluation, notifications, and performance tracking.

2.3 Functional Requirements

Functional requirements define the core operations and functionalities of the system. These requirements describe what the system should do to meet user expectations.

2.3.1 User Management

- Users should be able to register using basic details
- Secure login and logout functionality must be provided
- Password protection and authentication should be ensured

2.3.2 Course Management

- Users can view available courses
- Ability to select and enroll in courses
- Display of course content such as notes, videos, and materials

2.3.3 Quiz Module

- Users can attempt quizzes after completing topics
- Timer-based quiz functionality should be available
- Questions should be auto-generated or pre-defined
- System should submit quiz automatically when time ends

2.3.4 Result and Feedback System

- Automatic evaluation of answers
- Instant display of results
- Feedback generation for improvement

2.3.5 Coding Assistance (CodeXpert)

- Provide coding practice questions
- Generate code suggestions
- Help in debugging errors

2.3.6 Admin Features

- Admin can manage users and courses

- Ability to update or delete content
- Monitor user performance and activities

2.4 Non-Functional Requirements

Non-functional requirements define the quality attributes of the system and how it performs under different conditions.

2.4.1 Performance

The system should provide quick responses and handle multiple users simultaneously without delay.

2.4.2 Usability

The user interface should be simple, intuitive, and easy to navigate for all types of users.

2.4.3 Security

User data must be protected using authentication and encryption techniques to prevent unauthorized access.

2.4.4 Scalability

The system should be able to handle an increasing number of users and data without affecting performance.

2.4.5 Reliability

The platform should work consistently without crashes or errors.

2.4.6 Availability

The system should be available 24/7 for users with minimal downtime.

2.5 Hardware Requirements

The hardware requirements for the system are minimal since it is a web-based platform.

- A computer or laptop
- Minimum 4GB RAM (8GB recommended for better performance)
- Stable internet connection
- Cloud server or local server for hosting

2.6 Software Requirements

The system requires the following software tools and technologies:

- **Operating System:** Windows, Linux, or macOS
- **Web Browser:** Google Chrome, Microsoft Edge
- **Workflow Tool:** n8n (for automation)
- **Programming Languages:** JavaScript / Python
- **AI Tools:** ChatGPT, Gemini AI, Notion AI, NotebookLM
- **Database:** Firebase / MySQL (for storing data)

2.7 System Architecture

The system follows a layered architecture to ensure flexibility and scalability.

Frontend Layer

This layer provides the user interface where users interact with the system. It includes login pages, course dashboards, and quiz interfaces.

Backend Layer

The backend handles data processing, user requests, and business logic. It ensures smooth communication between frontend and database.

AI Layer

This layer integrates:

- Generative AI for content creation
- CodeXpert for coding assistance
- Agentic AI for automation

Database Layer

Stores user information, course details, quiz data, and results securely.

Workflow Automation (n8n)

n8n connects all components and automates tasks such as:

- Quiz evaluation
- Notifications
- Data processing

2.8 Feasibility Analysis

Feasibility analysis helps determine whether the project can be successfully implemented.

2.8.1 Technical Feasibility

The project is technically feasible as it uses existing tools and technologies that are widely available and easy to integrate.

2.8.2 Economic Feasibility

The system is cost-effective because most tools used are free or low-cost, reducing overall development expenses.

2.8.3 Operational Feasibility

The system is easy to use and requires minimal training, making it suitable for students and administrators.

2.9 Use Case Analysis

Use case analysis describes how different users interact with the system.

Student Use Cases

- Register and login
- Select courses
- Access learning materials
- Take quizzes with timer
- View results and feedback

Admin Use Cases

- Manage users
- Add or update courses
- Monitor student performance

2.10 Advantages of the System

- Provides personalized learning experience
- Saves time through automation
- Improves coding skills using AI

- Easy to use and accessible
- Enhances student engagement
- The system provides a **personalized learning experience** by adapting content based on user performance and preferences.
- It improves **learning efficiency** by offering instant feedback, helping students understand mistakes quickly.
- Integration of **Generative AI** helps in generating study materials, explanations, and quizzes automatically.
- **CodeXpert support** enhances programming skills by providing real-time coding assistance and debugging help.
- **Agentic AI automation** reduces manual effort by handling tasks like evaluation, notifications, and workflow management.
- The platform is **user-friendly**, making it easy for students and beginners to navigate and use effectively.
- It saves time for both students and instructors through **automation of repetitive tasks**.
- The system is **accessible anytime and anywhere**, allowing flexible learning without time restrictions.
- It supports **interactive learning**, increasing student engagement through quizzes and practical activities.
- The platform is **scalable**, meaning it can handle more users and courses as the system grows.
- It reduces dependency on traditional teaching methods by providing **smart digital learning solutions**.
- Continuous tracking of performance helps students to **monitor their progress and improve regularly**.

2.11 Limitations of the System

- Requires internet connectivity
 - AI-generated content may need verification
 - Limited customization in initial stages
- ✧ The system is highly dependent on a stable internet connection, and performance may be affected in low-network conditions.

- ✧ AI-generated content (from Generative AI and CodeXpert) may sometimes produce incorrect or irrelevant outputs, requiring manual verification.
- ✧ The system may face scalability challenges if a large number of users access it simultaneously without proper server support.
- ✧ Initial setup and integration of AI tools (like n8n workflows and APIs) can be complex and time-consuming.
- ✧ Limited personalization in early stages, as the system may require more data to provide accurate recommendations.
- ✧ Data privacy and security concerns may arise if proper protection mechanisms are not implemented.
- ✧ The system may not fully replace human teachers, especially for complex explanations and emotional support.
- ✧ Maintenance and updates are required regularly to keep the system running efficiently and up to date.
- ✧ Some advanced features may require paid APIs or subscriptions, increasing the overall cost.
- ✧ Compatibility issues may occur on older devices or browsers.
- ✧ Debugging AI-generated code can be difficult for beginners if the logic is not clearly understood.
- ✧ The system depends on third-party tools (AI platforms), so any downtime or changes in those services can affect performance.

CHAPTER 3

SYSTEM DESIGN

3.1 Introduction

The system design of the proposed project focuses on developing an intelligent and user-friendly **AI-based EdTech Learning Platform**. This platform is designed by integrating advanced technologies such as **Generative AI, CodeXpert, and Agentic AI** along with **n8n workflow automation**. The purpose of the system design is to define how different components of the system interact with each other to provide efficient learning, automated processes, and real-time feedback to users.

The design ensures that the system is scalable, reliable, and capable of handling multiple users while delivering personalized learning experiences.

3.2 System Architecture

The system follows a layered architecture that includes frontend, backend, AI layer, database, and automation workflow.

- The **frontend layer** provides an interface for users to interact with the system, such as login, course selection, and quizzes.
- The **backend layer** processes user requests and manages system operations.
- The **AI layer** integrates Generative AI for content creation, CodeXpert for coding assistance, and Agentic AI for automation and decision-making.
- The **database layer** stores user data, course details, quiz questions, and results.
- The **n8n workflow** connects all components and automates tasks like evaluation and notifications.

This architecture ensures smooth communication between all parts of the system.

3.3 n8n Workflow Design

n8n plays a key role in automating the system processes. It acts as the backbone of the project by connecting different modules and handling workflows efficiently.

The workflow starts when a user submits input (such as quiz answers). The data is received through a webhook and processed using function nodes. Then, AI services evaluate the input and generate results. Based on conditions, the system stores the results in the database and sends feedback to the user.

This automation reduces manual effort and improves system efficiency.

3.4 Module Design

The system is divided into different modules for better organization and functionality.

- The **User Module** handles registration, login, and user management.
- The **Course Module** allows users to browse and select courses.
- The **Learning Module** provides study materials generated using AI.
- The **Quiz Module** conducts quizzes with a timer and manages user responses.
- The **Evaluation Module** checks answers automatically and generates scores.
- The **CodeXpert Module** supports coding practice by suggesting code and debugging errors.
- The **Agentic AI Module** automates decision-making and tracks user progress.

Each module works together to provide a complete learning experience.

3.5 Data Flow of the System

The data flow in the system follows a structured process. First, the user logs into the system and selects a course. The system then provides learning content, after which the user attempts a quiz.

The quiz answers are sent to the backend through the n8n workflow, where AI evaluates the responses. The results are stored in the database, and feedback is displayed to the user. This continuous flow ensures smooth interaction and real-time updates.

3.6 Database Design

The database is designed to store all necessary information in an organized manner. It includes tables for users, courses, quizzes, and results.

- The **user table** stores login and profile details.
- The **course table** contains course information and content.
- The **quiz table** stores questions and answers.
- The **result table** records scores and feedback.

This structured design helps in efficient data storage and retrieval.

3.7 User Interface Design

The user interface is designed to be simple, interactive, and easy to use. It includes pages such as login, dashboard, course selection, quiz interface, and result display.

The design focuses on providing a smooth user experience with clear navigation and responsive layouts, ensuring that users can easily access all features without confusion.

3.8 System Workflow

The overall workflow of the system is straightforward and efficient. The user registers and logs into the platform, selects a course, and studies the provided content. After learning, the user takes a quiz with a timer.

The system automatically evaluates the answers using AI, stores the results, and displays feedback. Agentic AI helps in suggesting the next steps based on user performance, making the system more intelligent and adaptive.

AI-BASED WORKFLOW AUTOMATION SYSTEM SYSTEM DESIGN

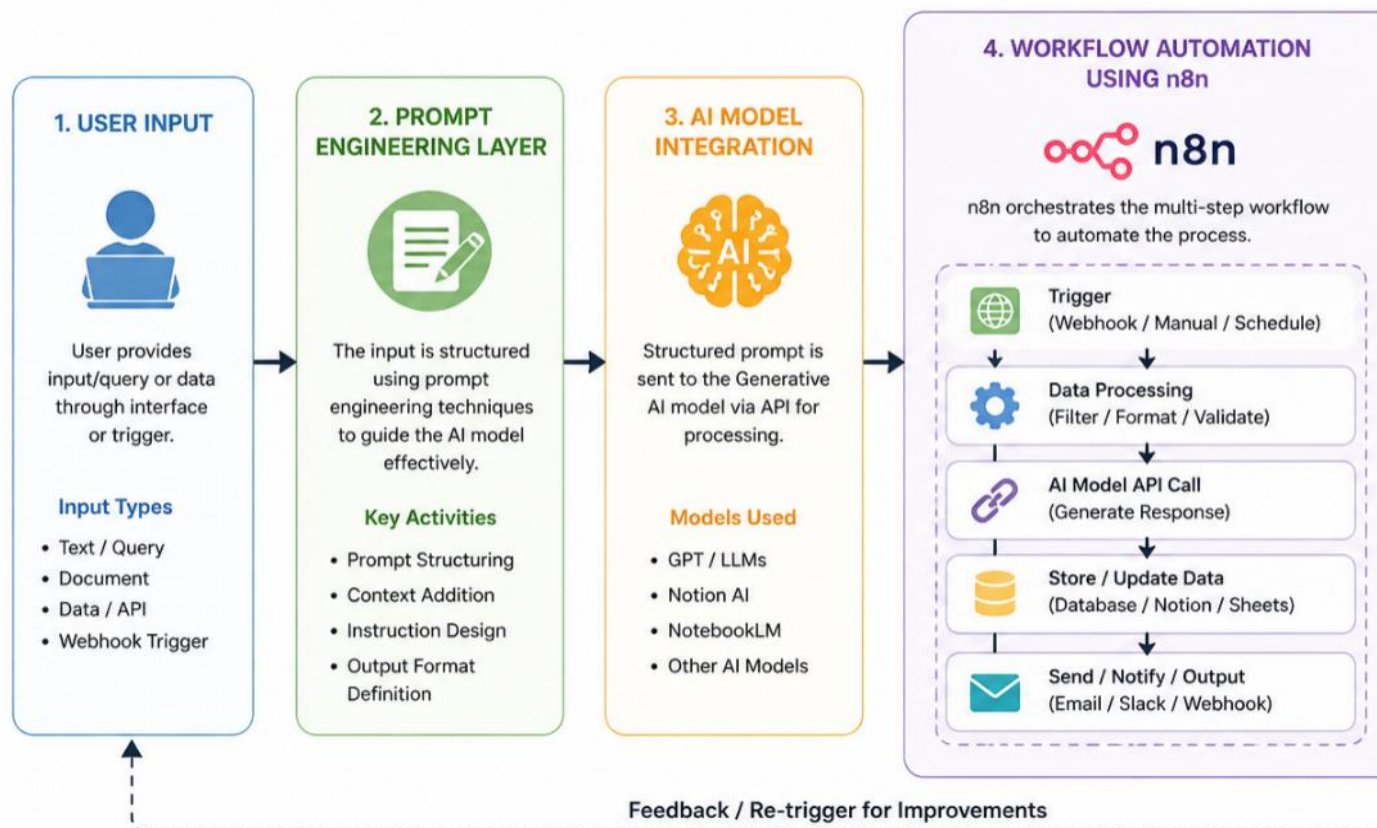


Fig 4.1: AI-Based Workflow Automation System Architecture

CHAPTER 4

IMPLEMENTATION

The implementation plan describes the step-by-step process of developing and deploying the proposed **AI-based EdTech Learning Platform**. It focuses on how the system is built using **Generative AI, CodeXpert, Agentic AI, and n8n workflow automation**. This plan ensures that the development process is organized, efficient, and completed within a structured timeline.

The implementation is divided into multiple phases, starting from requirement analysis to final deployment and testing. Each phase plays an important role in building a reliable and functional system.

4.2 Implementation Phases

The development of the system is carried out in the following phases:

4.2.1 Requirement Analysis Phase

In this phase, all system requirements are identified and analyzed. The features such as user login, course selection, quiz module, and AI integration are clearly defined. This phase helps in understanding the scope and objectives of the project.

4.2.2 System Design Phase

After analyzing requirements, the system design is prepared. The architecture, modules, database structure, and workflows are designed. This phase ensures that all components are properly planned before development begins.

4.2.3 Development Phase

In this phase, the actual coding and implementation of the system take place. The frontend interface is developed for user interaction, and backend logic is implemented for processing

data. AI tools are integrated to provide intelligent features such as content generation and coding assistance.

4.2.4 Integration Phase

All modules are integrated into a single system. The n8n workflow is configured to connect different components and automate tasks such as quiz evaluation and notifications. This phase ensures smooth communication between all parts of the system.

4.2.5 Testing Phase

The system is tested to identify and fix errors. Functional testing, performance testing, and usability testing are performed to ensure that the system works correctly and efficiently.

4.2.6 Deployment Phase

After successful testing, the system is deployed on a server or cloud platform. Users can access the platform through a web browser.

4.2.7 Maintenance Phase

In this phase, the system is monitored and updated regularly. Bugs are fixed, and new features can be added to improve performance and user experience.

4.3 Tools and Technologies Used

The implementation of the project uses the following tools and technologies:

- **n8n:** For workflow automation
- **Generative AI Tools:** ChatGPT, Gemini AI for content generation
- **CodeXpert:** For coding assistance and debugging
- **Frontend Technologies:** HTML, CSS, JavaScript
- **Backend Technologies:** Node.js / Python (if required)
- **Database:** Firebase / MySQL
- **Web Browser:** Chrome / Edge

These tools help in building an efficient and intelligent system.

4.4 Step-by-Step Implementation of the System

The implementation of the system follows these steps:

1. Set up the development environment and install required tools
2. Create user interface for login and course selection
3. Develop backend logic for handling user requests
4. Design and create database for storing data
5. Integrate Generative AI for content and quiz generation
6. Implement CodeXpert module for coding support
7. Configure n8n workflow for automation
8. Develop quiz module with timer functionality
9. Implement evaluation system for automatic scoring
10. Test the system and fix errors
11. Deploy the system for user access

4.5 Role of n8n in Implementation

n8n plays a major role in the implementation process by automating workflows. It connects different modules such as user input, AI processing, and database storage.

For example:

- When a user submits quiz answers, n8n processes the input
- AI evaluates the answers
- Results are stored automatically
- Feedback is sent to the user

This reduces manual work and improves system efficiency.

4.6 Challenges in Implementation

Some challenges faced during implementation include:

- Integrating multiple AI tools into a single system
- Managing data flow between modules
- Handling errors in AI-generated outputs

CHAPTER 5

ADVANTAGES AND DISADVANTAGES

Advantages

- The system provides a **smart and personalized learning experience** using AI technologies, helping students understand concepts better.
- It saves time by **automating tasks** such as quiz evaluation, feedback generation, and content creation.
- **Generative AI** helps in creating study materials, explanations, and quizzes quickly.
- **CodeXpert module** improves coding skills by giving real-time suggestions and debugging support.
- **Agentic AI** enables intelligent decision-making, such as recommending next learning steps based on performance.
- The platform is **user-friendly and easy to access**, allowing students to learn anytime and anywhere.
- It increases **student engagement** through interactive quizzes and instant feedback.

Disadvantages

- The system is highly dependent on a **stable internet connection** for proper functioning.
- AI-generated content may sometimes be **inaccurate or misleading**, requiring verification.
- Initial setup and integration of tools like **n8n and AI APIs** can be complex.
- The system may face **performance issues** if many users access it simultaneously.
- Some advanced features may require **paid AI services**, increasing cost.
- Users may become **over-dependent on AI**, reducing critical thinking skills.
- Data privacy and security risks may arise if proper protection is not implemented.

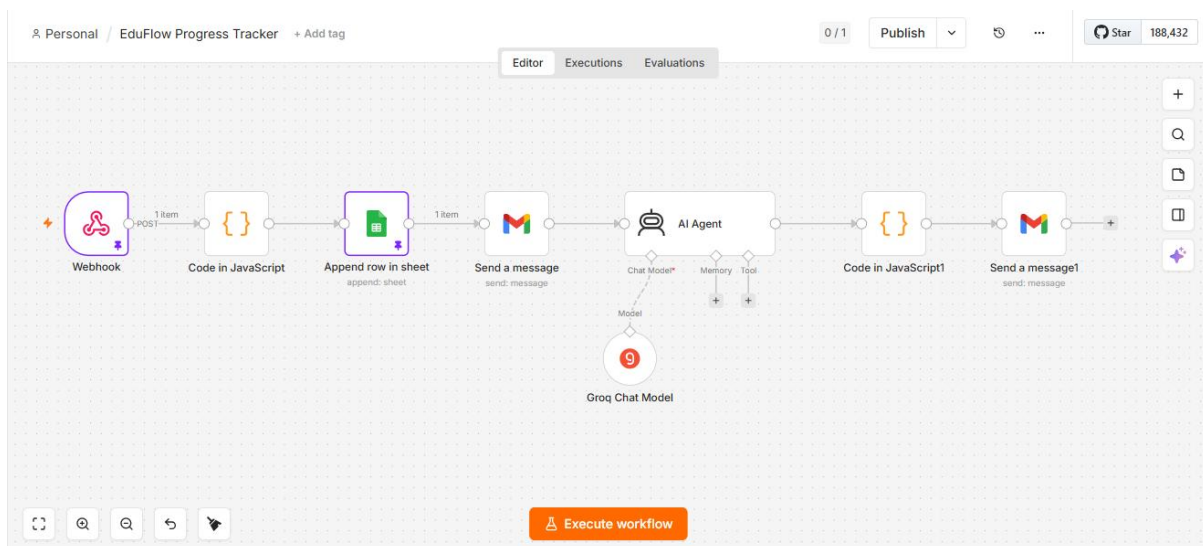
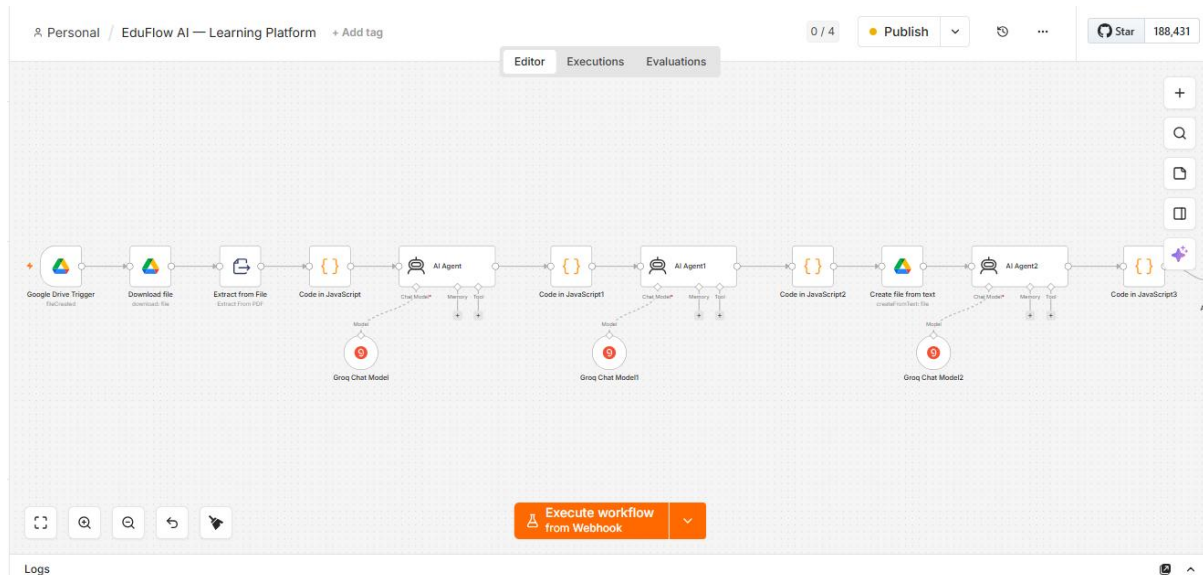
CHAPTER 6

APPLICATIONS

- ✧ The system can be widely used in **schools and colleges** to enhance traditional teaching methods by integrating AI-based learning and automated assessment tools.
- ✧ It is highly suitable for **online education platforms**, allowing students to access courses, AI-generated study materials, and quizzes from anywhere at any time.
- ✧ The project supports **self-paced learning**, enabling individuals to learn new skills, especially programming and technical subjects, without depending on instructors.
- ✧ It can be implemented in **coding learning platforms**, where the CodeXpert module provides real-time coding assistance, debugging support, and solution suggestions.
- ✧ The system is beneficial for **coaching centers and training institutes** to manage courses, conduct tests, and track student performance efficiently.
- ✧ It can be used in **corporate training environments** to train employees, assess their knowledge, and provide AI-based feedback for improvement.
- ✧ The platform enables **personalized and adaptive learning**, where content and difficulty levels are adjusted based on the learner's performance using Agentic AI.
- ✧ It is useful in **remote and distance learning systems**, helping students continue education without physical presence, especially in situations like online classes or hybrid learning models.

CHAPTER 7

SCREENSHOTS AND RESULTS



EduFlow Quiz Bank												
File Edit View Insert Format Data Tools Extensions Help												
100% 123 Default... 10 B I A												
question_number												
number	subject	chapter	type	question	option_a	option_b	option_c	option_d	answer	marks	explanation	
1	Science	Cell Biology	mcq	What is the primary function of the nucleus?	To provide structural support	To regulate gene expression	To produce energy	To synthesize proteins	B	2	The nucleus is the control center of the cell.	
2	Science	Cell Biology	true_false	The nucleus is the control center of the cell.					TRUE	2	The nucleus is the control center of the cell.	
3	Science	Cell Biology	fill_blank	The mitochondria are the _____ of the cell, responsible for generating energy.					powerhouses	2	Mitochondria are the powerhouses of the cell.	
4	Science	Cell Biology	mcq	Which organelle is responsible for the synthesis of proteins?	Mitochondria	Ribosomes	Lysosomes	Golgi apparatus	B	2	Ribosomes are involved in the synthesis of proteins.	
5	Science	Cell Biology	true_false	Lysosomes are involved in the synthesis of proteins.					FALSE	2	Lysosomes are involved in the degradation of macromolecules.	
6	Science	Cell Biology	fill_blank	The process of cellular respiration occurs in the _____ and is responsible for generating energy.					mitochondria	2	Cellular respiration occurs in the mitochondria.	
7	Science	Cell Biology	mcq	What is the function of the Golgi apparatus?	To synthesize proteins	To package and transport materials	To generate energy	To digest and recycle materials	B	2	The Golgi apparatus is responsible for packaging and transporting materials.	
8	Science	Cell Biology	mcq	Which of the following is a true statement about cells?	They have a nucleus	They have a cell wall	They lack a nucleus	They have a cell membrane	C	2	Prokaryotic cells lack a nucleus.	
9	Science	Cell Biology	true_false	The endoplasmic reticulum is involved in the synthesis of proteins.					TRUE	2	The endoplasmic reticulum is involved in the synthesis of proteins.	
10	Science	Cell Biology	mcq	What is the function of the cytoskeleton?	To provide structural support	To regulate gene expression	To produce energy	To synthesize proteins	A	2	The cytoskeleton provides structural support and is involved in cell movement.	
11	Computer Science	Introduction to Programming	mcq	What is the primary function of a compiler?	To execute code	To debug code	To translate code	To design code	C	2	A compiler translates source code into machine code.	
12	Computer Science	Introduction to Programming	true_false	Python is a statically-typed language.					FALSE	2	Python is a dynamically-typed language.	
13	Computer Science	Introduction to Programming	fill_blank	The CPU is responsible for executing instructions and controlling the flow of data within the computer.					brain	2	The CPU is the brain of the computer.	
14	Computer Science	Introduction to Programming	mcq	What is the purpose of a loop?	To skip over code	To repeat a set of instructions	To store data	To input data	B	2	A loop allows you to repeat a set of instructions.	
15	Computer Science	Introduction to Programming	true_false	JavaScript is a server-side programming language.					FALSE	2	JavaScript is a client-side programming language.	
16	Computer Science	Introduction to Programming	fill_blank	The process of finding and fixing errors in code is called _____.					debugging	2	Debugging is the process of finding and fixing errors in code.	
17	Computer Science	Introduction to Programming	mcq	What is the difference between a variable and a constant?	#ERROR!	#ERROR!	#ERROR!	#ERROR!	A	2	A variable can change its value, while a constant cannot.	
18	Computer Science	Introduction to Programming	mcq	What is the purpose of a function?	To store a constant	To store a change	To perform a calculation	To input data	B	2	A function is a block of code that performs a specific task.	
19	Computer Science	Introduction to Programming	true_false	HTML is a programming language.					FALSE	2	HTML is a markup language.	

Student Progress							
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CONCLUSION

The proposed AI-based EdTech learning platform successfully integrates **Generative AI, CodeXpert, and Agentic AI** to create a smart and interactive learning environment. The system enhances the overall learning experience by providing personalized content, automated quiz evaluation, and real-time feedback.

By using AI technologies, the platform reduces manual effort for both students and educators while improving learning efficiency and engagement. The integration of tools like n8n workflow automation ensures smooth execution of processes such as data handling, evaluation, and result generation.

The project also supports self-learning, skill development, and adaptive education, making it highly useful in modern digital learning environments. Although there are some limitations such as dependency on internet connectivity and AI accuracy, the overall system proves to be effective, scalable, and innovative.

In conclusion, this project demonstrates how AI can transform traditional education into a more intelligent, efficient, and user-friendly system, providing better outcomes for learners.

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